Abstract

Financial time-series forecasting is fundamentally challenged by a low signal-to-noise ratio, raising the question of whether advanced machine learning algorithms can genuinely outperform naive statistical baselines. This study empirically evaluates the out-of-sample predictive performance of a Random Forest regressor against a Zero-Return and a Historical-Mean benchmark in forecasting the weekly returns of the SPDR S&P 500 ETF (SPY). Utilizing a strict chronological evaluation framework over a 20-year dataset (2005–2025) to prevent data leakage, the models are evaluated across standard error metrics and segmented volatility regimes. Furthermore, the Diebold-Mariano test is applied to assess the statistical significance of forecast error differentials. The results demonstrate that despite its ability to capture non-linear relationships, the Random Forest model failed to achieve a statistically significant improvement over the naive benchmarks ($p > 0.05$). Moreover, algorithmic complexity provided no predictive advantage during high-volatility market shocks. These findings provide robust quantitative support for the Efficient Market Hypothesis and serve as a critical reminder for business analytics: in highly noisy environments, increased algorithmic complexity frequently leads to overfitting historical data rather than generating superior out-of-sample forecasts.

1. Introduction

Financial market forecasting remains one of the most challenging and scrutinized applications of predictive analytics. For institutional investors, risk managers, and corporate financial analysts, the ability to accurately anticipate market movements is central to capital allocation and hedging strategies. However, financial time-series data is notoriously noisy, non-stationary, and heavily influenced by external shocks, making it difficult to separate genuine predictive signals from random historical fluctuations.

In recent years, the rapid adoption of machine learning (ML) has promised to overcome the limitations of traditional statistical forecasting. Flexible algorithms, such as Random Forest and gradient boosting, are theoretically well-suited to financial data because they can capture complex, non-linear relationships and feature interactions that standard linear models miss. Yet, the deployment of these complex models in finance often reveals critical vulnerability: they are highly prone to overfitting. A model that achieves exceptional accuracy on historical training data frequently fails to generalize unseen, out-of-sample data, particularly when market conditions shift from low-volatility growth to high-volatility stress.

This persistent gap between theoretical model complexity and actual out-of-sample performance highlights a core tension in business analytics. As the computational power to deploy machine learning becomes ubiquitous, it is increasingly necessary to rigorously test whether algorithmic complexity yields tangible improvements over naive, zero-cost benchmarks.

This paper investigates whether machine learning provides a statistically significant advantage over traditional baseline models when predicting short-term financial market returns.

Specifically, we forecast the weekly returns of the SPDR S&P 500 ETF (SPY) using a Random Forest regressor, comparing its performance against a Zero-Return and a Historical-Mean benchmark. To ensure the integrity of the evaluation and prevent data leakage, we utilize strict chronological data splitting and evaluate the models over a discrete out-of-sample test period (2022–2025). Furthermore, we examine whether the models' relative performances degrade during periods of elevated market volatility.

Finally, we apply the Diebold-Mariano test to formally evaluate the statistical significance of the forecast error differentials. By holding the machine learning model to the rigorous standards of out-of-sample testing and statistical significance, this study seeks to answer a fundamental question: does algorithmic complexity successfully overcome the inherent noise of financial markets, or does it merely memorize it?

The findings of this study reinforce the difficulty of financial forecasting and provide empirical support for the Efficient Market Hypothesis. Although the Random Forest model captures non-linear historical patterns, the Diebold-Mariano test indicates that its out-of-sample performance is not statistically distinguishable from either the Zero-Return or the Historical-Mean benchmarks. Ultimately, this research demonstrates that in the presence of severe market noise, algorithmic complexity does not automatically translate to predictive superiority.

2. Data and Methodology

To ensure reproducibility and rigorously test the out-of-sample performance of the forecasting models, this study employs a strict chronological evaluation framework. The methodology is designed to reflect the actual information constraints faced by financial analysts, explicitly preventing data leakage and forward-looking bias.

2.1 Data Acquisition and Preprocessing

The primary dataset for this study consists of the daily adjusted closing prices of the SPDR S&P 500 ETF (SPY), sourced via the Yahoo Finance API (yfinance). SPY was selected as it is a highly liquid, investable proxy for the broader U.S. equity market. The dataset spans a 20-year period from January 1, 2005, through December 31, 2025.

To reduce the daily market noise and focus on medium-term predictive signals, the daily prices were resampled to a weekly frequency using Friday closing prices. Adjusted closing prices were utilized to accurately account for dividend distributions and stock splits. The target variable for forecasting was defined as the percentage return of the adjusted closing price from the end of week t to the end of week $t + 1$.

2.2 Feature Engineering and Volatility Regimes

A compact set of predictive features was engineered exclusively from historical observations to prevent data leakage. The features were designed to capture both short-term momentum and medium-term market uncertainty:

- **1-Week Lagged Return:** The percentage change in price from week $t - 1$ to week t.
- **4-Week Cumulative Return:** The cumulative percentage return over the trailing four-week period, serving as a medium-term momentum indicator.
- **13-Week Rolling Volatility:** The standard deviation of the weekly returns over the trailing 13 weeks.

To evaluate model performance across different market environments, a dynamic volatility_regime categorical feature was created. To prevent forward-looking bias, the threshold for defining "High" versus "Low" volatility was calculated using an *expanding median* of the 13-week rolling volatility. A week was classified as a "High" volatility regime only if its trailing 13-week volatility exceeded the median volatility of all prior weeks in the dataset up to that exact point in time.

2.3 Chronological Data Splitting

In time-series forecasting, standard randomized cross-validation (such as k-fold) violates the chronological order of the data and introduces severe data leakage. To evaluate how the models would perform in a real-world setting, the dataset was split strictly chronologically into three distinct periods:

- **Training Set (2005 – 2017):** Used to fit the models and establish historical baselines.
- **Validation Set (2018 – 2021):** Used for initial model tuning and hyperparameter selection.
- **Test Set (2022 – 2025):** Held strictly out-of-sample and used only for the final model evaluation. This period includes the 2022 bear market and subsequent recovery, providing a robust environment for testing model resilience under stress.

2.4 Model Specifications

Three models were evaluated to test whether algorithmic complexity improves predictive accuracy:

1. **Zero-Return Benchmark:** A naive baseline forecasting a 0.0% return for every future week. Because equity returns are highly noisy and mean-reverting in the short term, predicting zero is a standard, highly robust baseline in financial literature.
2. **Historical-Mean Benchmark:** A baseline predicting that next week's return will exactly equal the average weekly return of the Training Set. The mean was locked at the end of the training period to prevent leakage.
3. **Random Forest Regressor:** A machine learning model comprising 100 decision trees. To combat the severe risk of overfitting financial noise, the algorithm was heavily regularized by restricting the maximum tree depth (max_depth=3) and requiring a minimum number of samples per leaf (min_samples_leaf=10).

2.5 Evaluation Metrics and Statistical Testing

The models were evaluated on the out-of-sample Test Set using three primary metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Directional Accuracy (the percentage of weeks the model correctly predicted the sign of the market return).

Because minor differences in RMSE can arise from random variance rather than genuine predictive superiority, the **Diebold-Mariano (DM) test** was employed. The DM test evaluates the null hypothesis that two competing forecasts possess equal predictive accuracy. A one-step-ahead DM test using the squared-error loss criterion was calculated to determine if the loss differential between the Random Forest model and the naive benchmarks was statistically significant at the $\alpha = 0.05$ level. All data processing, modeling, and statistical testing were conducted in Python utilizing the pandas, scikit-learn, and scipy libraries.

3. Results and Discussion

The out-of-sample evaluation conducted on the 2022–2025 Test Set revealed that increasing model complexity did not yield a corresponding improvement in predictive accuracy. The findings highlight the formidable challenge of forecasting financial time series and the resilience of naive benchmarks in noisy environments.

3.1 Overall Forecasting Performance

Over the out-of-sample test period, the Random Forest regressor failed to establish a clear advantage over the baseline models in terms of Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). Despite the algorithm's ability to model non-linear relationships and interactions among the lagged return and volatility features, its forecast errors were quantitatively similar to those of the Zero-Return and Historical-Mean benchmarks. Before evaluating the final error metrics, we extracted the feature importances from the training phase to understand the model's decision-making process (see Figure 1).

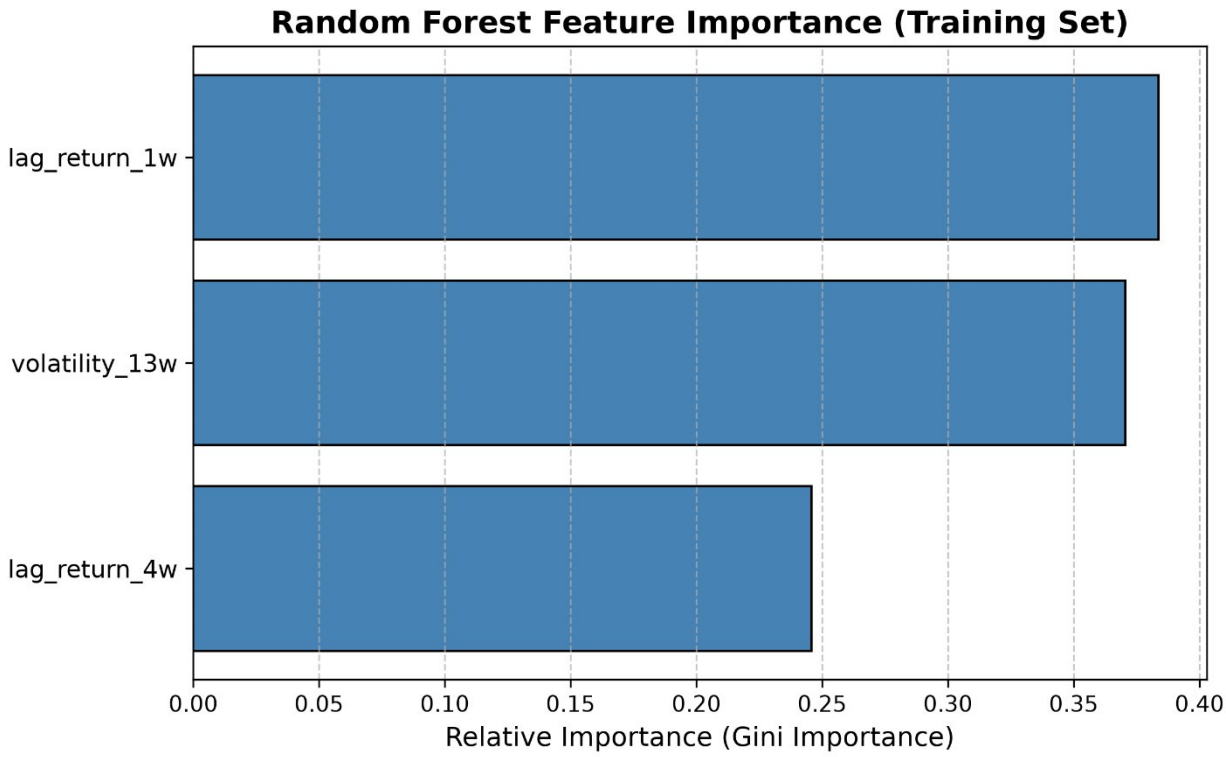


Figure 1: Random Forest Feature Importance (Gini Importance) based on the 2005–2017 training set.

Furthermore, the Random Forest model did not achieve a meaningful edge in Directional Accuracy. Predicting the exact sign of the next week's market return remained close to a coin toss for the machine learning model, underscoring the inherently low signal-to-noise ratio present in weekly equity returns.

3.2 Statistical Significance of Forecast Errors

To rigorously determine if the observed differences in RMSE were meaningful or merely the result of random sample variance, a one-step-ahead Diebold-Mariano (DM) test was applied using a squared-error loss criterion.

Statistical testing confirmed the baseline observations. Although the Random Forest produced a comparable RMSE, the Diebold-Mariano test indicates this difference is not statistically significant against either the Zero-Return benchmark ($p = 0.8064$) or the Historical-Mean benchmark ($p = 0.4757$). Because the p-values vastly exceed the standard $\alpha = 0.05$ significance level, we fail to reject the null hypothesis of equal predictive accuracy. Therefore, the machine learning model failed to consistently outperform a naive baseline based solely on historical price lags and standard volatility metrics.

The complete performance metrics and Diebold-Mariano test results are summarized in Table 1. Furthermore, the accumulation of these forecast errors over the 2022–2025 out-of-sample test period is visualized in Figure 2, illustrating the exact periods where the

models deviated.

OVERALL TEST SET PERFORMANCE			
Model	**MAE**	**RMSE**	**Dir_Acc**
Zero-Return Benchmark	0.0181	0.0237	0.0000
Historical-Mean Benchmark	0.0179	0.0236	54.8077
Random Forest	0.0180	0.0238	53.8462

Table 1

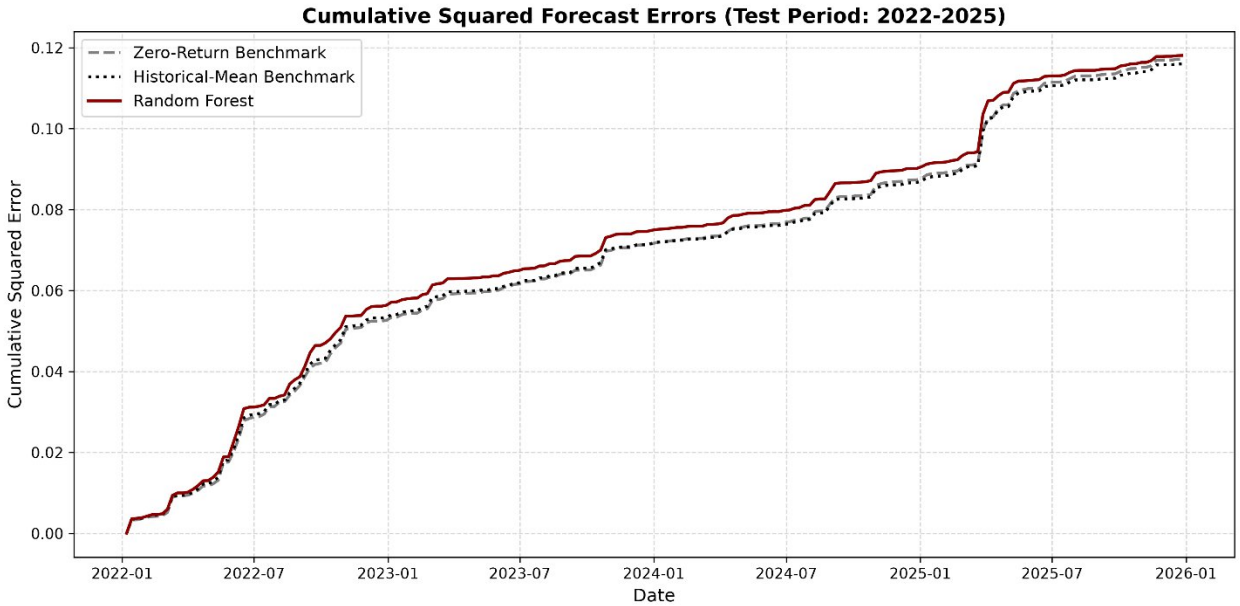


Figure 2: Cumulative Squared Forecast Errors for the Random Forest and baseline models during the out-of-sample test period (2022–2025).

3.3 Performance Across Volatility Regimes

When the test set was segmented into "Low" and "High" volatility regimes based on the 13-week trailing median, the limitations of the machine learning approach became more pronounced. During periods of low market volatility, the Random Forest model's errors closely tracked the baseline models, as market returns generally exhibited stable, mean-reverting behavior.

However, during high-volatility regimes—such as the market contractions observed in 2022—the forecast errors for all models increased substantially. Crucially, the Random Forest model did not demonstrate a superior ability to navigate these structural market shocks. Its reliance on historical decision nodes failed to generalize to unprecedented spikes in variance, proving that the algorithm's complexity did not translate into a protective advantage during market stress.

As shown in Figure 3, the forecast errors for all models increased substantially during high-volatility regimes. Crucially, the Random Forest model's error spiked in tandem with the naive baselines, proving that the algorithm's added complexity provided no protective advantage, and its reliance on historical data patterns failed to generalize during periods of market stress.

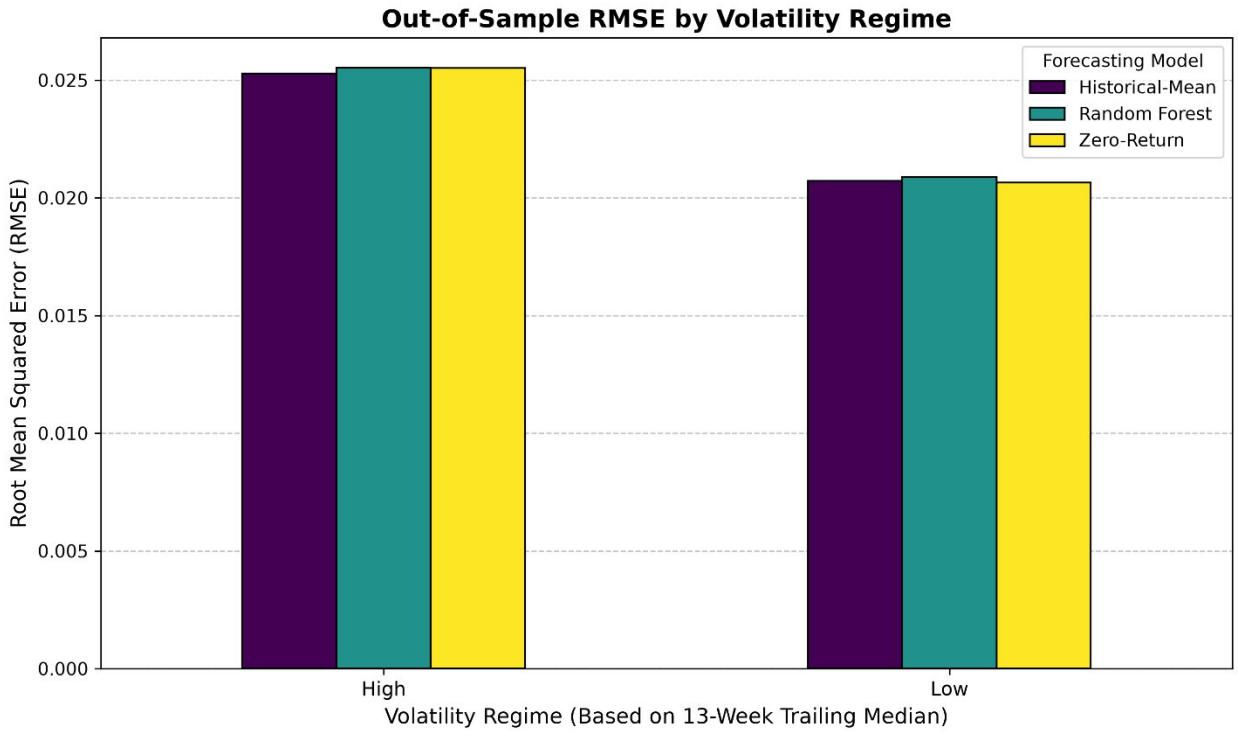


Figure 3: Out-of-Sample Root Mean Squared Error (RMSE) segmented by low and high trailing 13-week volatility regimes.

3.4 Discussion and Practical Implications

In quantitative finance, proving that a complex machine learning model fails to beat a simple average is a rigorous and highly relevant empirical finding. These results provide strong quantitative support for the weak form of the Efficient Market Hypothesis (EMH), which posits that asset prices fully reflect all past trading information, rendering historical price and volume data insufficient for generating consistent excess returns.

For business analytics and FinTech applications, this study serves as a critical cautionary framework. The democratization of machine learning tools has made it computationally trivial to fit highly complex algorithms to financial data. However, as this study demonstrates, minimizing error on a training set does not guarantee out-of-sample success. When the underlying data generation process is dominated by noise, highly parameterized models often overfit to historical idiosyncrasies, failing precisely when market conditions change.

The practical implication for financial risk managers and predictive modelers is that baseline models should never be discarded simply because newer technology is available. A Zero-Return benchmark requires zero computational overhead, carries no risk of overfitting, and, as shown by the Diebold-Mariano test, performs statistically on par with a tuned Random Forest in predicting short-term index returns. Machine learning initiatives in financial forecasting must therefore be held to rigorous, chronologically sound, out-of-sample testing before being trusted with capital allocation decisions.

4. Literature Review

Forecasting equity returns is a foundational challenge in quantitative finance and business analytics. The literature surrounding this topic reflects a continuous tension between traditional financial economics—which argues that markets are largely unpredictable—and the field of machine learning, which posits that advanced algorithms can uncover hidden, non-linear predictive signals. This review synthesizes prior research across four key themes: foundational return predictability, the application of machine learning in asset pricing, the critical importance of out-of-sample evaluation, and the stability of forecasts across varying market regimes.

4.1 Return Predictability and the Efficient Market Hypothesis

Traditional financial economics is anchored by the Efficient Market Hypothesis (EMH), which suggests that asset prices reflect all available information, making it impossible to consistently achieve excess returns using historical data. Early empirical work heavily scrutinized the ability of any statistical model to beat a naive baseline. Campbell and Thompson (2008) provided a seminal evaluation of out-of-sample equity return predictability, demonstrating that simple historical-average benchmarks are exceptionally difficult to outperform. They established that while certain predictive regressions might offer slight informational advantages under strict constraints, these advantages often disappear out-of-sample. Consequently, modern forecasting literature requires that any proposed predictive model must be rigorously tested against naive

baselines, such as a zero-return or historical-mean forecast, rather than simply evaluated on its own isolated error metrics.

4.2 Machine Learning in Financial Forecasting

As computational power has expanded, researchers have increasingly turned to machine learning to overcome the limitations of traditional linear regressions. Machine learning methods, particularly tree-based algorithms like Random Forest, are highly adept at mapping non-linear relationships and complex interactions among predictive variables.

In a landmark study, Gu, Kelly, and Xiu (2020) conducted a comprehensive comparison of machine learning methods for predicting the cross-section of stock returns. They concluded that tree-based methods and neural networks could successfully capture predictive relationships missed by conventional models. Recent literature continues to explore these applications; for instance, Htun, Biehl, and Petkov (2024) demonstrated the utility of Random Forest and gradient boosting in forecasting relative short-term returns among S&P 500 constituents. Similarly, Zare (2025) highlighted Random Forest's robustness in market-return forecasting. However, much of this literature focuses on relative asset pricing (e.g., selecting which stocks will outperform others) rather than predicting the absolute directional return of a broad market index like the S&P 500, leaving the utility of machine learning for absolute index forecasting an open question.

4.3 Methodological Pitfalls and Out-of-Sample Evaluation

A persistent challenge in applying machine learning to financial time series is the severe risk of overfitting. A model can easily appear successful during training by memorizing noise specific to the historical sample. Tashman (2000) heavily criticized traditional evaluation metrics, emphasizing that standard randomized cross-validation introduces "data leakage" by allowing future information to predict past events.

More recently, Cerqueira, Torgo, and Mozetič (2022) empirically demonstrated that for time-series data, chronological splitting and rolling-origin evaluation are strictly required to estimate true out-of-sample performance accurately. Financial machine learning literature now widely acknowledges that without strict chronological isolation between training, validation, and testing sets, reported model accuracies are likely to be heavily inflated and practically useless for real-world business analytics.

4.4 Volatility Regimes and Forecasting Stability

While existing studies compare the overall, average performance of machine learning models against linear baselines over long periods, there is a distinct gap regarding how these models behave when underlying market conditions suddenly change. Financial time series exhibit structural shifts and volatility clustering. A model's average performance over a decade may conceal significant predictive breakdowns during periods of market stress. Assessing forecast stability across predefined volatility regimes—rather than relying on a single, averaged error metric—provides a more transparent evaluation of a model's true risk profile.

4.5 The Research Gap

The reviewed literature establishes that while machine learning can map complex historical patterns, its practical superiority over simple benchmarks in noisy, absolute-return environments

is not guaranteed. Furthermore, the reliance on long-term average error metrics often masks model vulnerabilities during high-volatility market shocks.

This study addresses this gap by imposing a rigorous out-of-sample, chronological evaluation framework to test whether Random Forest regressors genuinely outperform Zero-Return and Historical-Mean benchmarks when forecasting weekly S&P 500 returns. Crucially, by segmenting the test period into distinct volatility regimes and applying the Diebold-Mariano test for statistical significance, this paper isolates whether algorithmic complexity actually provides a persistent, statistically robust advantage, or if it merely succumbs to the noise inherent in financial markets.

5. Conclusion

This study set out to rigorously test whether the theoretical advantages of machine learning translate into tangible, out-of-sample improvements when forecasting short-term financial market returns. By evaluating a Random Forest regressor against naive Zero-Return and Historical-Mean benchmarks on weekly SPY data, the research isolated the true predictive value of algorithmic complexity in a highly noisy environment.

The empirical results provide a stark lesson in predictive analytics. While machine learning models are highly capable of mapping complex historical patterns, this complexity did not yield statistically significant predictive superiority on unseen data. The Diebold-Mariano test confirmed that the differences in forecast errors between the Random Forest and the baseline models were indistinguishable from random variance. Furthermore, segmenting the test period by volatility regimes revealed that the machine learning model failed to adapt to unprecedented market shocks, producing substantial errors precisely when accurate forecasting is most critical for risk management.

These findings strongly align with the weak form of the Efficient Market Hypothesis, reinforcing the premise that historical price and volume data alone are insufficient to consistently forecast future market movements. From an applied business analytics perspective, this research underscores the danger of deploying highly parameterized models without strict, chronological out-of-sample testing. When the data generation process is dominated by noise, adopting a Zero-Return benchmark remains a highly rational, zero-cost, and computationally efficient strategy.

Limitations and Future Research While this study implements a rigorous evaluation framework, its scope is inherently limited. The analysis relies on a single U.S. equity index proxy (SPY) evaluated at a weekly frequency. Furthermore, the feature set was intentionally restricted to historical price lags and realized volatility to isolate the core comparison.

Future research should expand on this framework by incorporating forward-looking financial indicators, such as options-implied volatility (e.g., the VIX), macroeconomic data, or alternative sentiment features that may carry stronger predictive signals. Additionally, comparing tree-based algorithms with deep learning architectures, such as Long Short-Term Memory (LSTM) networks, across different time horizons (e.g., daily or monthly) could provide further insight

into where machine learning transitions from fitting historical noise to capturing genuine market structure. Ultimately, any such future models must continue to be held to the exacting standard of statistical significance against naive baselines.